Ge 158 Physical Principles of Remote Sensing
Problem Set 2: Oblique incidence, layered media, and surface and volume scattering

Notation and conventions (as in the lecture notes): Time-harmonic fields use $e^{j\omega t}$ with $j^2 = -1$, so the complex dielectric constant is $\varepsilon = \varepsilon' - j\varepsilon''$ with $\varepsilon'' \geq 0$. A tilde denotes a phasor, $\mathbf{E}(x, y, z, t) = \mathrm{Re}[\tilde{\mathbf{E}}(x, y, z)\, e^{j\omega t}]$. All media are nonmagnetic. Use $c = 3 \times 10^8$ m/s, $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m, and $\mu_0 = 4\pi \times 10^{-7}$ H/m. Several problems ask for plots; submit the plots along with your code.

AI use: Include a brief statement of how you used AI tools.

Problem 1: A uniform plane wave in air ($\varepsilon_1 = 1$), which occupies the half-space $z \geq 0$, has the electric field phasor

$$\tilde{\mathbf{E}}^i = \hat{\mathbf{y}}\, 20\, e^{-j(3x-4z)} \qquad (\mathrm{V/m}), \tag{1}$$

where x and z are in meters. The wave is incident on the planar surface $z = 0$ of a lossless dielectric with $\varepsilon_2 = 4$ that occupies the half-space $z \leq 0$. Draw a ray diagram (as in Lecture 5) showing the incident, reflected, and transmitted wave vectors and their electric and magnetic fields, and determine:

a. Whether the incident wave is H or V polarized;

b. The angular frequency ω and the angles of incidence, reflection, and transmission;

c. The Fresnel reflection and transmission coefficients;

d. Time-domain expressions for the reflected electric and magnetic fields;

e. The reflectivity Γ and transmissivity $\mathbb{T}$, and show that $\Gamma + \mathbb{T} = 1$.

Time-domain expressions are those that contain the term ωt.

Problem 2: Imagine that an offshore oil rig has caught fire and sunk, rupturing the pipe that carries oil from the ocean floor to the rig. Oil is gushing from the broken pipe, and a large slick is spreading across the ocean surface. A remote sensing instrument that operates at 20 GHz is available to help with cleanup and containment. At this frequency, the double-Debye model of Lecture 1 gives the dielectric constant of seawater (salinity 35 psu, $T = 20$ °C) as $\varepsilon_3 = 34.7 - j36.3$; the dielectric constant of crude oil is $\varepsilon_2 = 2.1 - j0.1$; and air has $\varepsilon_1 = 1$. Model the slick as a layer of uniform thickness d overlying a seawater half-space, with perfectly smooth, horizontal surfaces, and use the effective reflection coefficient ρ_{eff} of a layer over a half-space from Lecture 6.

a. Plot $|\rho_{\mathrm{eff}}|$ for H and V polarization (on the same plot) versus d from 0 to 30 mm at normal incidence ($\theta_1 = 0$).

b. Repeat for an incidence angle $\theta_1 = 55°$.

c. Most of your curves should show a clear lobe structure. In your own words, explain what causes the lobes. One curve shows essentially no lobes; identify it and explain why.

d. From your plot for part (a), estimate the thickness at which the first minimum occurs and the spacing between successive minima. Compare these with the wavelength λ_2 in the oil, and explain the relationship.

The angle $\tilde{\theta}_2$ in the oil is complex because the oil is lossy (Lecture 5). When you compute $\cos\tilde{\theta}_2$ (and $\cos\tilde{\theta}_3$ in the seawater), choose the root that gives a wave that decays as it propagates downward.

Problem 3: Estimating soil moisture is a classic application of radar remote sensing. Here we explore a simple approach for inferring the volumetric soil moisture m_v, the dimensionless ratio of water volume to total volume. A 1-GHz radar observes a bare loam soil with a randomly rough, horizontal surface and measures the co-polarized normalized radar cross sections

$$\sigma^0_{hh} = -25.57 \text{ dB}, \qquad \sigma^0_{vv} = -20.63 \text{ dB}, \tag{2}$$

where dB indicates decibels, $\sigma^0_{pp}\,[\text{dB}] = 10\log_{10}\sigma^0_{pp}$. For the type of soil in the study area, the surface roughness has a Gaussian correlation function with correlation length $\ell = 10$ cm. The Bragg wavenumber of the observations is $K_{\text{Bragg}} = 30$ rad/m. Assume the soil is lossless, with relative permittivity

$$\varepsilon' = 3.25 + 2.2m_v + 138.7m_v^2, \qquad 0 \le m_v \le 0.5, \tag{3}$$

and use the first-order small-perturbation model from Lecture 8.

a. What is the incidence angle θ_i of the observations? Carry at least four significant figures through the rest of the problem; the inversion is sensitive to θ_i.

b. Show that the ratio $\sigma^0_{hh}/\sigma^0_{vv}$ does not depend on the RMS height h or the correlation length ℓ. Use the ratio to estimate m_v.

c. Using your value of m_v, estimate the RMS height h of the surface. Calculate kh, where k is the radar wavenumber, and comment on whether the small-perturbation model is appropriate for this surface.

d. Before calculating, write down how you expect σ^0_{hh} and σ^0_{vv} to change with increasing incidence angle. Then, for your values of m_v and h, plot σ^0_{hh} and σ^0_{vv} (in dB) versus θ_i for $20^\circ \le \theta_i \le 70^\circ$. Note that K_{Bragg} changes with θ_i; the value of 30 rad/m applies only to the measurements above. Does the plot look like what you expected? Explain any differences.

Problem 4: Now suppose a crop grows on the field from Problem 3. Keep the same radar, incidence angle, and soil (with your values of m_v and h), and model the crop as a uniform canopy of height d filled with isotropic scatterers. For the canopy, take the illustrative values $\kappa_{\text{e}} = 0.1$ Np/m for the extinction coefficient and $\kappa_{\text{s}} = 0.01$ Np/m for the scattering coefficient (for both polarizations). From Lecture 9, the co-polarized backscatter for isotropic scatterers is

$$\sigma^0_{pp} = \underbrace{\Upsilon^2\sigma^0_{\text{s},pp}}_{\text{ground}} + \underbrace{\frac{\kappa_{\text{s}}}{\kappa_{\text{e}}}\,\frac{\cos\theta_i}{2}\left(1-\Upsilon^2\right)\left(1+(\Gamma^{\text{coh}}_p)^2\Upsilon^2\right)}_{\text{volume}} + \underbrace{4\kappa_{\text{s}}\,d\,\Gamma^{\text{coh}}_p\,\Upsilon^2}_{\text{canopy–ground}}, \tag{4}$$

where $\Upsilon = e^{-\kappa_{\text{e}}d/\cos\theta_i}$ is the transmissivity of the canopy, $\sigma^0_{\text{s},pp}$ is the backscatter from the bare soil (from your small-perturbation model in Problem 3), and $\Gamma^{\text{coh}}_p = \Gamma_p\, e^{-4k^2h^2\cos^2\theta_i}$ is the coherent reflectivity of the soil surface, where $\Gamma_p = |\rho_p|^2$ is the Fresnel reflectivity of the soil for p polarization (Lecture 5).

a. For each co-polarization (hh and vv), plot the total σ^0_{pp} and each of the three contributions (in dB) versus d for $0.01 \le d \le 3$ m (the volume and canopy–ground terms vanish at $d = 0$). For each polarization, find the canopy height at which the ground contribution equals the sum of the other two contributions.

b. Calculate the change in σ^0_{vv} (in dB) when m_v increases from 0.1 to 0.3 (keeping h fixed, but recalculating ε, Γ_p, and $\sigma^0_{\mathrm{s},pp}$ at each m_v) for bare soil and for canopies with $d = 1$ m and $d = 2$ m. What does this imply for estimating soil moisture beneath vegetation? How would you expect your answers to change for a radar operating at a higher frequency? (A qualitative answer is fine.)

Problem 5: Practice working with remote sensing data. Sign in to the Google Earth Engine Code Editor (code.earthengine.google.com), create a new script, and paste in the code below. Earth Engine requires a registered (noncommercial) Cloud project; set yours up well before the due date, and let us know right away if you have trouble.

The script makes a map centered on Iceland with three layers over the standard Google basemap (switch to the satellite view to see the terrain). Use the "Layers" menu at the top right of the map to turn layers on and off and to adjust their transparency. The two grayscale layers show the median VV-polarized normalized radar cross section σ^0_{vv} (in dB) measured by the Sentinel-1 synthetic aperture radar (C band, 5.4 GHz) in winter and in summer (read the script to see the dates). The Sentinel-1 data in Earth Engine are already in dB, so do not take the logarithm again. The red–white–blue layer shows their difference, winter minus summer: red means winter backscatter is higher, and blue means summer backscatter is higher. Because dB is logarithmic, the difference in dB is the ratio of the two cross sections, expressed in dB.

a. Submit a screenshot of the difference layer.

b. Where do you see the largest differences in backscatter between winter and summer? Using what we have learned in class (dielectric properties, Lecture 1; reflection and penetration, Lectures 2 and 4–6; surface and volume scattering, Lectures 7–9), explain the likely cause of the change.

c. Find a region where the backscatter changes little between seasons, and one where summer backscatter is higher than winter backscatter. Speculate on the physical reasons for each.

There are many reasonable answers; we are looking for thoughtful explanations grounded in the physics from class.

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// Region and point of interest (Iceland)
var iceland = ee.Geometry.Rectangle([-24.6, 63.2, -13.4, 66.6]);
var center = ee.Geometry.Point([-18.81, 64.81]);

// Sentinel-1 VV backscatter (dB), IW mode, ascending passes over Iceland
var s1 = ee.ImageCollection('COPERNICUS/S1_GRD')
    .filterBounds(iceland)
    .filter(ee.Filter.eq('instrumentMode', 'IW'))
    .filter(ee.Filter.listContains('transmitterReceiverPolarisation', 'VV'))
    .filter(ee.Filter.eq('orbitProperties_pass', 'ASCENDING'))
    .select('VV');

// Median backscatter in winter and summer, and their difference
var vvWinter = s1.filterDate('2024-01-01', '2024-03-01').median();
var vvSummer = s1.filterDate('2024-07-01', '2024-09-01').median();
var vvDiff = vvWinter.subtract(vvSummer);

Map.centerObject(center, 6.5);
Map.addLayer(vvWinter, {min: -25, max: 0}, 'VV winter (dB)');
Map.addLayer(vvSummer, {min: -25, max: 0}, 'VV summer (dB)');
Map.addLayer(vvDiff, {min: -10, max: 10, palette: ['blue', 'white', 'red']},
             'VV winter minus summer (dB)');
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